# Negative Exponents

2026-10-26

**Rules to remember**

$$a^{-n} = \frac{1}{a^n}$$

$$\frac{1}{a^{-n}} = a^n$$

$$a^m \cdot a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

**Example: simplify and write with positive exponents**

$$\frac{x^3}{x^{-2}} \cdot x^{-7}$$

$$= x^{3-(-2)} \cdot x^{-7}$$

$$= x^5 \cdot x^{-7}$$

$$= x^{-2}$$

$$= \frac{1}{x^2}$$

**Rewrite using only positive exponents.**

**1.** $7^{-2}$

Answer: _____

**2.** $x^{-5}$

Answer: _____

**3.** $\dfrac{1}{4^{-3}}$

Answer: _____

**4.** $3y^{-4}$

Answer: _____

**5.** $a^{-2}b^3$

Answer: _____

**Rewrite using a negative exponent.**

**6.** $\frac{1}{6^4}$

Answer: _____

**7.** $\frac{1}{m^2}$

Answer: _____

**8.** $2^5$

Answer: _____

**Evaluate. Write your answer as a whole number or a fraction.**

**9.** $2^{-3}$

Answer: _____

**10.** $10^{-2}$

Answer: _____

**11.** $(-3)^{-2}$

Answer: _____

**12.** $\frac{1}{5^{-2}}$

Answer: _____

**Simplify. Write your answer using only positive exponents.**

**13.** $x^5 \cdot x^{-3}$

Answer: _____

**14.** $4^{-6} \cdot 4^2$

Answer: _____

**15.** $a^{-2} \cdot a^{-5}$

Answer: _____

**16.** $\dfrac{y^3}{y^7}$

Answer: _____

**17.** $\dfrac{x^2}{x^{-4}}$

Answer: _____

**18.** $\dfrac{n^{-3}}{n^{-8}}$

Answer: _____

**19.** $(x^{-2})^3$

Answer: _____

**20.** $(m^{-3})^{-2}$

Answer: _____